

SOLVE THE 8 QUEENS PROBLEM

PYTHON CODE:

8 Queens Problem - Print First 7 Solutions

N = 8

count = 0

```
def print_board(board):
    for row in board:
        print(" ".join("Q" if cell == 1 else "." for cell in row))
    print()
```

```
def is_safe(board, row, col):
```

```
    for i in range(col):
        if board[row][i] == 1:
            return False
```

```
    i, j = row, col
    while i >= 0 and j >= 0:
        if board[i][j] == 1:
            return False
        i -= 1
        j -= 1
```

```
    i, j = row, col
    while i < N and j >= 0:
        if board[i][j] == 1:
            return False
        i += 1
        j -= 1
```

```
    return True
```

```
def solve(board, col):
    global count
```

```

if count == 7:
    return True

if col == N:
    count += 1
    print(f"Solution {count}:")
    print_board(board)
    return False

for row in range(N):
    if is_safe(board, row, col):
        board[row][col] = 1
        if solve(board, col + 1):
            return True
        board[row][col] = 0

return False

```

```

board = [[0] * N for _ in range(N)]
solve(board, 0)
print("True")

```

OUTPUT:

Solution 1:

```

Q .....
..... Q .
.... Q ...
..... Q
. Q .....
... Q ....
..... Q ..
.. Q .....

```

Solution 2:

```

Q .....
..... Q .
... Q ....
..... Q ..
..... Q

```

. Q
... Q
.. Q

Solution 3:

Q
... Q
..... Q
.. Q
..... Q
... Q
. Q
..... Q

Solution 4:

Q
... Q
..... Q
..... Q
.. Q
..... Q
. Q
... Q

Solution 5:

..... Q
Q
... Q
. Q
..... Q
.. Q
..... Q
... Q

Solution 6:

... Q
Q
... Q
..... Q
. Q

..... Q .
.. Q
..... Q ..

Solution 7:

..... Q
Q
..... Q
... Q
. Q
..... Q .
.. Q
..... Q ..

True